Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Stable Quantum Dots with Tunable Infrared Bandgaps			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Bio and Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	Optical Materials	Therapeutic Devices & Diagnostic Devices	Nuclear Medicine and Molecular Imaging Diagnostic Devices
Concept of Proposed Technology	The Smith Lab at UIUC develops semiconductor nanocrystals (SNCs) with tunable absorption and light emission throughout the infrared (IR) spectrum. SNCs are a breakthrough photonic material that provide vivid new colors for displays such as QLED TVs, and the field is now mature for visible-spectrum SNCs. However, it is anticipated that their biggest impact will come from their ability to absorb and emit light across spectral bands inaccessible to traditional materials, especially the IR. Detectors, cameras, LEDs, lasers, and contrast agents are far less efficient and more expensive in the IR than their extensively developed counterparts at visible wavelengths. Low-cost photonic applications in the IR could allow high-resolution visualization of biological structures deep in the body, rapid identification of materials and gases, and atmospheric signal transmission beyond the limits of the visible spectrum. However, wavelengths longer than ~1 μm require expensive and bulky cooling modules due to vibrational coupling to electronic transitions. The Smith Lab recently made advances in IR-based SNCs with figures of merit matching those of the best visible spectrum SNCs throughout the short-wave and mid-wave IR. Through this collaboration, we aim to apply these in next-generation IR optoelectronic devices functioning at room temperature.			

Rationale for Joint R&D Project	We are looking for partners in the development of integrated devices applying quantum dots as either optically active components of sensors (photodiodes, cameras), light sources (LEDs, lasers), and as molecular contrast agents. Our lab has expertise in engineering, production, and scaling of quantum dots and other nanocrystals with bandgaps in the infrared from the short-wave region to the mid-wave region where devices are expensive and bulky and require cooling modules.
Keywords	1. Infrared 2. Photoluminescence 3. Sensor 4. In vivo imaging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (6)	End of the Project	Level (8)
R&D Objectives	The Smith Lab develops and produces semiconductor nanocrystals (SNCs) with tunable bandgaps and photophysical properties. Specifically for this project, we intend to produce high quality SNCs with bandgaps in the short-wave infrared (SWIR) spectrum (1–3 μm) and mid-wave infrared (MWIR) spectrum (3–5 μm). We recently advanced a new class of colloidal SNCs based on HgCdSe, HgSe which have received little attention in comparison with closely related II-VI family members of CdSe and CdTe (the prototypical colloidal quantum dots) and HgTe (used in MWIR photodetectors). This is a major advance because the HgSe SNCs are derived from high-quality CdSe SNC precursors, functioning as sacrificial templates via interdiffusion-enhanced cation exchange which we developed 2 years ago. Because CdSe SNCs are highly advanced, through IE-CE, HgSe and HgCdSe SNCs can now be tuned over extremely broad sizes, shapes, and composite structures to tune bandgaps across the SWIR and MWIR. At the same time, we have advanced a new class of SWIR SNCs with no heavy metal atoms composed of Ag_2S which is expected to allow deep imaging as contrast agents for biomedical applications, complete metabolism, and full excretion in conventional time windows for radiologic imaging without the need for radioisotopes.			
R&D Target Specifications	• Photodiodes functioning at SWIR and MWIR with room temperature detectivity $>10^{12}$ Jones up to 5 μm wavelength. • MWIR camera and the first demonstration of MWIR QD fluorescence imaging up to 3 μm.			
R&D Plan	The goal is to advance IR SNC technologies from TRL 6 to TRL 8 through integrated materials development, device fabrication, and system-level validation. The project will focus on short-wave infrared (SWIR, 1–3 μm) and mid-wave infrared (MWIR, 3–5 μm) operation at room temperature, addressing key limitations of current IR optoelectronic devices. **Aim 1: Advanced Nanocrystal Synthesis and Optimization (TRL 6 to TRL 7)** The research institute will optimize HgSe, HgCdSe, and Ag_2S SNCs using interdiffusion-enhanced cation exchange (IE-CE) and related scalable colloidal synthesis methods. Key activities include: • Precise bandgap tuning across SWIR and MWIR • Optimization of surface chemistry and shelling (CdS and CdSe) for room-temperature carrier lifetime • Optimization of quantum yield and metabolic pathways of heavy-metal-free Ag_2S SNCs for biomedical imaging applications Milestones: reproducible synthesis protocols for scaling up to 1 μmol batch sizes, batch-to-batch consistency, and material stability suitable for device integration without inert gas requirements. **Aim 2: Device Integration and Prototype Development (TRL 7)** In collaboration with the Korean SME partner, SNCs will be integrated into functional optoelectronic devices for: • SWIR and MWIR photodiodes with room-temperature detectivity $>10^{12}$ Jones up to 5 μm • Prototype MWIR imaging arrays and proof-of-concept MWIR fluorescence imaging systems Milestones: thin-film deposition onto sensors and sensor arrays and performance characteristics including responsivity, noise, and detectivity across SWIR and MWIR bands. **Aim 3: System Validation and Reliability Testing (TRL 8)** Prototype devices will undergo performance and reliability testing under application-relevant conditions. Biomedical imaging demonstrations will assess optical performance and suitability for in vivo contrast in live rodents. Successful completion will position the technology for certification, standardization, and commercialization.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Smith Lab at UIUC will be responsible for the core scientific and technological development associated with the IR nanocrystals including • Design, synthesis, and characterization of SWIR and MWIR semiconductor nanocrystals (HgSe, HgCdSe, Ag_2S) and any core/shell composites. • Computational modeling of photophysical properties using density functional theory and effective mass approximation. • Development of scalable and reproducible synthesis and surface functionalization protocols. • Fundamental photophysical analysis, including absorption, emission, carrier dynamics, and stability. • Technical guidance for nanocrystal integration into devices, including material compatibility and interface engineering. • Training and knowledge transfer to the Korean SME partner through frequent communication. • Laboratory infrastructure, materials expertise, and scientific leadership.
Expected R&R of Korean SME Partner	The Smith Lab at UIUC has already worked extensively with Korean academic partners toward device integration (Young-Hoon Kim at Hanyang University) and successfully demonstrated $>10^{12}$ Jones detectivity at room temperature at 1.33 μm. We look forward to working with an additional SME partner at technological stages of productization and commercialization. The Korean SME partner is expected to lead applied development, system integration, and commercialization-oriented activities. Anticipated responsibilities include: • Device fabrication and engineering of photodiodes, imaging sensors, LEDs, or laser platforms incorporating IR SNC materials. • Development of scalable manufacturing and packaging processes compatible with industrial production. • Performance testing, reliability evaluation, and environmental qualification of prototype devices. • Market-driven optimization of device specifications based on target applications, especially sensing, imaging, biomedical diagnostics. • Personnel, fabrication facilities, and commercialization expertise. • Collaboration with UIUC for rapid translation of laboratory-scale materials to validated devices.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project aims to deliver next-generation IR optoelectronic technologies and devices that significantly reduce cost, complexity, and size compared with conventional IR systems with limited portability, potential for miniaturization due to the requirement for cryogenic cooling. Expected outcomes include: • High-performance SWIR and MWIR photodiodes operating at room temperature • First demonstrations of MWIR quantum dot fluorescence imaging up to 3 μm • Scalable, solution-processable IR nanocrystal materials suitable for commercial manufacturing • Heavy-metal-free IR contrast agents enabling new biomedical imaging modalities Commercial and industrial impacts include new products in IR sensing, imaging, and diagnostics, which have key applications in manufacturing, industrial inspection, and environmental monitoring through materials and gas identification, medical imaging, and advanced optoelectronics. For Korean SMEs, the collaboration provides access to proprietary IR nanomaterials and early-stage technologies that can be differentiated in global markets. Broader impacts will include reduced barriers to IR technology advancement and adoption, which are needed for semiconductors, advanced bio-technologies, and photonics. The collaboration also strengthens international R&D partnerships and accelerates the transition of academic breakthroughs into real-world applications.

6. References

(1) Lee, W. & Smith, A. M. Interdiffusion-enhanced cation exchange for HgSe and HgCdSe nanocrystals with infrared bandgaps. Nature Synthesis 3: 1243-1254 (2024).

(2) Lim, S. J., Schleife, A. & Smith, A. M. Optical determination of crystal phase in semiconductor nanocrystals. Nature Communications 8: 14849 (2017).

(3) Sarkar S, Arogundade OH, Cui Y, Hernandez Alvarez EI, Schleife A, Smith AM. Nanocrystal Synthesis with Alkoxy Reagents for Dispersion in Polar and Nonpolar Solvents. Nature Synthesis 4: 826-835 (2025).

(4) Sarkar S, Le P, Geng J, Liu Y, Han Z, Zahid MU, Nall D, Yeoan Y, Selvin PR, Smith AM. Short-wave infrared quantum dots with compact sizes as molecular probes for fluorescence microscopy. Journal of the American Chemical Society 142: 3449-3462 (2020).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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